

AI Forecasting Model: Global Oil Prices (2025-2030)

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Project Overview

This demo showcases a deep learning-powered Transformer model trained on global oil market data from 2010 to 2024. The goal is to forecast crude oil prices from 2025 to 2030 using historical variables such as:

- Crude Oil Prices (USD per barrel)
- Natural Gas Prices
- Global Oil Production (Million Barrels per Day)
- OECD Petroleum Stock Levels

The model was developed and deployed using Python, Darts, and Streamlit, and is hosted as a live web demo on Hugging Face.

Expo Use Case

This demo is part of Nayman AI & Data Consulting's exhibit at the World AI Technology Expo. Visitors can explore both a pre-trained forecast and run custom predictions with uploaded datasets. The app provides visualizations, annotations, and a downloadable summary.

Key Highlights

- Forecasting horizon: Jan 2025 to Dec 2030
- Model: Transformer (input chunk: 36 months, output: 60 months)
- Accuracy: Trends captured using covariates + attention-based architecture
- Platform: Streamlit + Hugging Face Spaces
- Supports live uploads and custom forecasting